A Discretize-First Linearized ADMM Solver for the Dynamic Schrödinger Bridge

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1 Introduction

This document describes the design and implementation of a 1D solver for the *dynamic Schrödinger Bridge* (SB) problem using a two-grid linearized Alternating Direction Method of Multipliers (ADMM). Three intertwined design choices shape the entire solver:

1. **Discretize first, then optimize.** The continuous PDE constraint is discretized on a grid before any optimality conditions are derived. ADMM is applied to the resulting finite-dimensional problem.

2. **Two staggered grids.** The objective and the constraint live most naturally on different grids. We maintain both explicitly and couple them via interpolation operators inside ADMM.

3. **Implicit diffusion discretization.** The diffusion term in the Fokker-Planck constraint is treated implicitly, rendering the scheme unconditionally stable and avoiding the parabolic CFL restriction $\Delta t = O(\Delta x^2)$.

Each choice is elaborated below after introducing the continuous problem.

2 The Continuous Problem

2.1 Benamou–Brenier Optimal Transport

The *Benamou–Brenier* (BB) formulation of optimal transport seeks a density–momentum pair (ρ, m) on the space-time cylinder $(t, x) \in [0, 1]^2$ that minimizes the kinetic energy

$$J(\rho, m) = \int_0^1 \int_0^1 \frac{1}{2} \frac{m(t,x)^2}{\rho(t,x)} \, dx \, dt \tag{1}$$

subject to the continuity equation with prescribed marginals,

$$\partial_t \rho + \partial_x m = 0, \qquad \rho(0,\cdot) = \rho_0, \quad \rho(1,\cdot) = \rho_1, \quad \partial_x \rho\big|_{x=0,1} = 0. \tag{2}$$

The integrand $m^2/(2\rho)$ is the kinetic energy density of a fluid with density ρ and momentum $m = \rho v$ where v is the velocity field.

2.2 The Schrödinger Bridge

Adding entropy regularization replaces the continuity equation with the *Fokker–Planck equation*,

$$\partial_t \rho + \partial_x m = \varepsilon\, \partial_{xx} \rho, \qquad \rho(0,\cdot) = \rho_0, \quad \rho(1,\cdot) = \rho_1, \tag{3}$$

with the same objective (1). The parameter $\varepsilon \geq 0$ controls the strength of diffusion:

- $\varepsilon = 0$: recovers the pure OT (Benamou–Brenier) problem.
- $\varepsilon > 0$: the Schrödinger Bridge; paths are smoother and more diffuse, corresponding to the most likely trajectory of N Brownian particles conditioned on their initial and final empirical distributions.

The term $m = \rho v$ retains its role as the momentum, but v is now a drift that guides the diffusive particles.

3 Discretize First, Then Optimize

3.1 Two Philosophies

A PDE-constrained optimization problem can be attacked in two orders:

Optimize-then-discretize (OtD): Derive the infinite-dimensional optimality conditions (Hamilton–Jacobi–Bellman equation, adjoint system, etc.) and only then discretize those PDEs for numerical solution. The resulting scheme is often non-trivially coupled and requires careful adjoint consistency.

Discretize-then-optimize (DtO): Directly discretize the state equation and the objective on a grid, producing a finite-dimensional constrained optimization problem. An off-the-shelf algorithm (here ADMM) is then applied to this discrete problem.

We adopt the **DtO** approach. The advantages in this context are:

- ADMM's convergence guarantees apply to the finite-dimensional problem without needing to track consistency of an adjoint PDE discretization.
- The prox and projection sub-steps are posed on a fixed grid and can be computed exactly or near-exactly at each iteration.
- The implicit diffusion treatment (Section 7) emerges naturally from the grid formulation rather than requiring separate analysis at the continuous level.

3.2 The Discrete Problem

Let N_t and N_x be the numbers of time and space cells, with $\Delta t = 1/N_t$ and $\Delta x = 1/N_x$. Cell centers are at $(t_{i+1/2}, x_{j+1/2})$ and cell edges (faces) are at integer nodes.

We discretize (ρ, m) on a space-time grid and require the discrete analogue of (3) to hold at every grid point. The objective is replaced by

$$J_h(\rho, m) = \Delta t\, \Delta x \sum_{i,j} \frac{m_{ij}^2}{2\,\rho_{ij}}, \tag{4}$$

with $J_h = +\infty$ whenever any $\rho_{ij} \leq 0$.

4 Two Grids and Why We Need Them

4.1 Motivation: Conflicting Requirements

The objective (4) requires ρ and m to live at the *same* spatial and temporal locations, so their ratio m^2/ρ can be formed pointwise.

The Fokker–Planck equation, on the other hand, is a conservation law of the form $\partial_t \rho + \partial_x m = \varepsilon \partial_{xx} \rho$. A *conservative* finite-difference discretization naturally lives on a staggered grid: the density ρ at cell centers, the flux m at cell faces. Using the same collocated grid for both ρ and m in the constraint produces a checkerboard-mode instability and requires artificial stabilization.

These two requirements are in tension:

- **Objective**: ρ and m collocated at cell centers.
- **Constraint**: ρ at integer time nodes $\times$ cell centers, m at half-integer times $\times$ cell faces.

4.2 Grid Definitions

We therefore maintain *two* sets of variables:

Grid 1 (collocated), variables (ρ, m): Both live at space-time cell centers $(t_{i+1/2}, x_{j+1/2})$.

$$\begin{aligned} \rho_{ij} &\approx \rho(t_{i+1/2},\, x_{j+1/2}), \quad \text{shape } N_t \times N_x, \\ m_{ij} &\approx m(t_{i+1/2},\, x_{j+1/2}), \quad \text{shape } N_t \times N_x. \end{aligned}$$

Grid 2 (staggered), variables $(\bar{\rho}, b)$: Density at integer times, flux at spatial faces.

$$\begin{aligned} \bar{\rho}_{ij} &\approx \rho(t_i,\, x_{j+1/2}), \quad \text{shape } (N_t - 1) \times N_x \quad \text{(interior time nodes; BCs absorbed into } d\text{)}, \\ b_{ij} &\approx m(t_{i+1/2},\, x_j), \quad \text{shape } N_t \times (N_x - 1) \quad \text{(interior faces; Neumann BCs enforced)}. \end{aligned}$$

The boundary values $\bar{\rho}^0 = \rho_0$ and $\bar{\rho}^{N_t} = \rho_1$ are not stored as unknowns; they appear on the right-hand side of the constraint.

4.3 Interpolation Operators

The two grids are linked by averaging operators:

$$(A_\rho\,\rho)_i = \tfrac{1}{2}(\rho_i + \rho_{i+1}), \quad i = 0, \ldots, N_t - 2, \tag{5}$$

$$(A_m\,m)_j = \tfrac{1}{2}(m_j + m_{j+1}), \quad j = 0, \ldots, N_x - 2. \tag{6}$$

Stacking these, the interpolation from Grid 1 to Grid 2 is written $A(\rho, m) = (A_\rho\,\rho,\ A_m\,m)$.

4.4 The ADMM Splitting

ADMM splits the problem by introducing duplicates:

$$\min_{(\rho,m),\,(\bar\rho,b)} J_h(\rho, m) + \iota_C(\bar\rho, b), \qquad \text{subject to} \quad A(\rho, m) = (\bar\rho, b), \tag{7}$$

where ι_C is the indicator of the Fokker–Planck constraint set

$$C = \big\{(\bar\rho, b) \mid (I_x \otimes D_t - \varepsilon L_x \otimes A_t)\bar\rho + (D_x \otimes I_t)\, b = d\big\}. \tag{8}$$

Each sub-problem is now tractable:

- The (ρ, m)-step minimizes J_h plus a quadratic penalty and decouples pointwise.
- The $(\bar\rho, b)$-step is a projection onto the affine constraint set C, which reduces to solving a single linear system.

5 Linearized ADMM

5.1 Why Linearization is Necessary

Standard (non-linearized) ADMM requires solving

$$(\rho^{k+1}, m^{k+1}) = \arg\min_{\rho,m} J_h(\rho, m) + \frac{\gamma}{2}\|A(\rho, m) - (\bar\rho^k, b^k) + \delta^k/\gamma\|^2. \tag{9}$$

Because A mixes neighboring values of ρ in time and neighboring values of m in space, this is a *coupled* minimization over the entire grid; the quadratic term in (9) cannot be solved pointwise.

5.2 The Linearization

Linearized ADMM replaces the $A(\rho, m)$ term inside the penalty with a first-order Taylor expansion around the current iterate, adding a proximal term with parameter $\tau > 0$:

$$(\rho^{k+1}, m^{k+1}) = \operatorname{prox}_{J_h/\tau}\left((\rho^k, m^k) - \frac{\gamma}{\tau}A^\top\left(A(\rho^k, m^k) - (\bar\rho^k, b^k) + \frac{\delta^k}{\gamma}\right)\right). \tag{10}$$

The argument to the proximal operator is an explicit function of the current iterate, so no coupled solve is needed. The proximal operator itself decouples pointwise (see Section 6).

Remark 1. *The stability condition for linearized ADMM is $\tau > \gamma\|A\|^2$. Since A_ρ and A_m are averaging operators, $\|A\|^2 \le 1$, so $\tau = 5\gamma$ is a safe, practical choice.*

5.3 The Three ADMM Steps

Let $\delta = (\delta_{\bar{\rho}}, \delta_b)$ be the dual variable (Lagrange multiplier for the ADMM constraint).

Step 1 — (ρ, m) proximal update. Compute the residual $r = A(\rho^k, m^k) - (\bar{\rho}^k, b^k) + \delta^k/\gamma$, form the gradient step

$$u = \rho^k - \frac{\gamma}{\tau} A_\rho^\top r_{\bar{\rho}}, \qquad v = m^k - \frac{\gamma}{\tau} A_m^\top r_b, \tag{11}$$

and then compute $(\rho^{k+1}, m^{k+1}) = \mathrm{prox}_{J_h/\tau}(u, v)$ pointwise (Section 6).

Step 2 — $(\bar{\rho}, b)$ projection update. With over-relaxation parameter $\alpha \in (1, 2)$:

$$(\bar{\rho}^{k+1}, b^{k+1}) = \mathrm{proj}_C\Big(\alpha\, A(\rho^{k+1}, m^{k+1}) + (1-\alpha)(\bar{\rho}^k, b^k) + \delta^k/\gamma\Big). \tag{12}$$

The projection solves a structured linear system (Section 8).

Step 3 — Dual update.

$$\delta^{k+1} = \delta^k + \gamma\big(\alpha\, A(\rho^{k+1}, m^{k+1}) + (1-\alpha)(\bar{\rho}^k, b^k) - (\bar{\rho}^{k+1}, b^{k+1})\big). \tag{13}$$

6 Pointwise Proximal Operator

For each cell (i, j), the proximal sub-problem is

$$(\rho^*, m^*) = \arg\min_{\rho \geq 0,\, m} \frac{\Delta t\, \Delta x}{2\tau} \frac{m^2}{\rho} + \frac{1}{2}(\rho - u)^2 + \frac{1}{2}(m - v)^2. \tag{14}$$

Setting $w = \Delta t\, \Delta x$, optimality in m gives

$$m^* = \frac{\tau\, \rho^*\, v}{\tau\, \rho^* + w/2}. \tag{15}$$

Substituting back into (14) and differentiating with respect to ρ yields a *cubic equation*:

$$\tau^2 \rho^3 - \tau^2 u\, \rho^2 + \tau\Big(w - \frac{w}{2}\Big)(2w - u\,\tau)\, \rho - uw^2/4 - \tfrac{w\tau}{2}\, v^2/4 = 0. \tag{16}$$

More precisely, the coefficients are

$$a_3 = \tau^2, \quad a_2 = \tau(2w - u\tau), \quad a_1 = w(w - 2u\tau), \quad a_0 = -uw^2 - \tfrac{w\tau}{2}\, v^2. \tag{17}$$

We take the **largest real root** $\rho^* > 0$ (following Papadakis et al. 2014). If $u \leq 0$ we set $\rho^* = 0$, $m^* = 0$, since the domain of J_h requires $\rho > 0$. A small positivity floor $\rho^* \leftarrow \max(\rho^*, 10^{-10})$ is applied after the solve to prevent division by zero in subsequent iterations.

The cubic is solved analytically (Cardano's method) in fully vectorized NumPy code, so there are no Python loops over grid points.

7 Implicit Diffusion and the CFL Constraint

7.1 The Parabolic CFL Condition

An *explicit* (forward-Euler) discretization of the diffusion term $\varepsilon\,\partial_{xx}\rho$ would give

$$\frac{\bar{\rho}^{n+1} - \bar{\rho}^n}{\Delta t} = \frac{\varepsilon}{\Delta x^2}(\bar{\rho}^n_{j+1} - 2\bar{\rho}^n_j + \bar{\rho}^n_{j-1}).$$

Von Neumann stability analysis requires the *parabolic Courant number* $\mu = \varepsilon\Delta t/\Delta x^2 \leq 1/2$. For fixed spatial resolution N_x this imposes

$$\Delta t \leq \frac{\Delta x^2}{2\varepsilon} = \frac{1}{2\varepsilon N_x^2}. \tag{18}$$

For the parameters $N_x = 128$, $\varepsilon = 0.05$ this gives $\Delta t \leq 6 \times 10^{-4}$, requiring $N_t \geq 1670$ time steps.

7.2 Our Implicit Treatment

The Fokker–Planck constraint (3) is discretized as

$$(I_x \otimes D_t - \varepsilon L_x \otimes A_t)\,\bar{\rho} + (D_x \otimes I_t)\,b = d, \tag{19}$$

where:

- D_t (shape $N_t \times (N_t - 1)$) is the first-difference operator in time, corresponding to the time derivative $\partial_t\rho$.
- A_t (shape $N_t \times (N_t - 1)$) is the time-averaging operator. The diffusion term is evaluated at the *average* of the density at adjacent integer-time nodes: this is an **implicit/semi-implicit (Crank-Nicolson-like) scheme**.
- $L_x = D_x D_x^\top$ is the negative Neumann Laplacian ($L_x \approx -\partial_{xx}$, positive semi-definite), so $+\varepsilon L_x$ correctly represents forward diffusion.
- $D_x \otimes I_t$ is the spatial divergence of the flux b.

Because A_t averages between two adjacent time levels, the diffusion term involves $\bar{\rho}$ at both t_i and t_{i+1}. This is equivalent to an implicit treatment of the diffusion, making the scheme **unconditionally stable** with respect to the diffusion coefficient. There is no restriction of the form (18).

7.3 Stiffness Ratio

A useful diagnostic is the *stiffness ratio*

$$\mu_h = \frac{\varepsilon}{\Delta x^2/\Delta t} = \frac{\varepsilon N_x^2}{N_t}. \tag{20}$$

When $\mu_h \gg 1$ the scheme is operating far outside the explicit stability region, yet convergence is maintained because the implicit diffusion enters the Fokker–Planck constraint solve (Step 2 of ADMM) rather than the time integration. In practice, values of μ_h up to ~ 10 are handled without difficulty.

Remark 2. *The advective CFL $|v|\Delta t/\Delta x \leq 1$ does not appear explicitly because the velocity $v = m/\rho$ is not discretized directly. Instead, m is the independent variable and the kinetic energy $m^2/(2\rho)$ is minimized directly.*

8 The Projection Step

8.1 KKT System

The projection $(\bar{\rho}^{k+1}, b^{k+1}) = \mathrm{proj}_C(p_{\bar{\rho}}, p_b)$ is the nearest point in C to $(p_{\bar{\rho}}, p_b)$. By KKT conditions, the solution satisfies

$$\bar{\rho}^{k+1} = p_{\bar{\rho}} - (I_x \otimes D_t - \varepsilon L_x \otimes A_t)^\top \phi, \tag{21}$$

$$b^{k+1} = p_b - (D_x \otimes I_t)^\top \phi \tag{22}$$

where the multiplier $\phi \in \mathbb{R}^{N_t \times N_x}$ solves the normal equations

$$M_{\mathrm{SB}}\, \phi = F, \tag{23}$$

with right-hand side $F = (I_x \otimes D_t - \varepsilon L_x \otimes A_t)\, p_{\bar{\rho}} + (D_x \otimes I_t)\, p_b - d$, and the symmetric positive semi-definite operator

$$\begin{aligned} M_{\mathrm{SB}} &= (I_x \otimes D_t - \varepsilon L_x \otimes A_t)(I_x \otimes D_t - \varepsilon L_x \otimes A_t)^\top + (D_x \otimes I_t)(D_x \otimes I_t)^\top \\ &= I_x \otimes D_t D_t^\top - \varepsilon L_x \otimes (A_t D_t^\top + D_t A_t^\top) + \varepsilon^2 L_x^2 \otimes A_t A_t^\top + D_x D_x^\top \otimes I_t. \end{aligned} \tag{24}$$

A direct assembly and factorization of M_{SB} would cost $O(N_t^3 N_x^3)$. We exploit its tensor-product structure for an $O(N_x N_t \log N_t)$ solve.

8.2 Efficient Solver via DCT and Tridiagonal Systems

The key observation is that $L_x = D_x D_x^\top$ is *diagonalized by the DCT-II basis* (Neumann boundary conditions). Applying DCT-II in space decouples (23) into N_x independent $N_t \times N_t$ tridiagonal systems, one per spatial mode $k = 0, \ldots, N_x - 1$:

$$T^k \hat{\phi}_k = \hat{F}_k, \tag{25}$$

where $\hat{F}_k$ is the k-th DCT-II mode of F and T^k is a symmetric tridiagonal matrix with entries

$$\text{main diagonal (interior):} \quad \alpha = \frac{2}{\Delta t^2} + \frac{\varepsilon^2 \lambda_k^2}{2} + \lambda_k, \tag{26}$$

$$\text{off-diagonal:} \quad \beta = -\frac{1}{\Delta t^2} + \frac{\varepsilon^2 \lambda_k^2}{4}, \tag{27}$$

$$\text{corner } (0,0): \quad \frac{1}{\Delta t^2} + \frac{\varepsilon \lambda_k}{\Delta t} + \frac{\varepsilon^2 \lambda_k^2}{4} + \lambda_k, \tag{28}$$

$$\text{corner } (N_t{-}1, N_t{-}1): \quad \frac{1}{\Delta t^2} - \frac{\varepsilon \lambda_k}{\Delta t} + \frac{\varepsilon^2 \lambda_k^2}{4} + \lambda_k, \tag{29}$$

where $\lambda_k = \frac{2}{\Delta x^2}(1 - \cos(\pi k/N_x)) \geq 0$ is the k-th eigenvalue of L_x.

For $k > 0$ (where $\lambda_k > 0$), T^k is strictly positive definite and is solved with SciPy's $O(N_t)$ banded solver. For $k = 0$ (the zero-mode of L_x), $T^0 = D_t D_t^\top$ is the Neumann Laplacian in time, which is singular; the right-hand side is guaranteed to be consistent by mass conservation, and the minimum-norm (mean-zero) solution is returned.

Computational cost. The DCT-II transform costs $O(N_x N_t \log N_x)$. Each of the N_x tridiagonal solves costs $O(N_t)$. The inverse DCT-II costs $O(N_x N_t \log N_x)$. The total is $O(N_x N_t(\log N_x + 1))$, which is essentially linear in the problem size $N_x N_t$.

9 Discrete Operators in Detail

9.1 Notation and Ordering

All 2D arrays use the convention: **time is the slow index** (row index) and **space is the fast index** (column index). Kronecker products are written with the time factor first: $B_t \otimes B_x$ acts on a vector with time-slow ordering.

9.2 One-Dimensional Operators

Table 1 summarizes the 1D building blocks.

Table 1: One-dimensional discrete operators.

Symbol	Shape	Action	Notes
D_t	$N_t \times (N_t - 1)$	Time first-difference	BCs in d
A_t	$N_t \times (N_t - 1)$	Time averaging	Semi-implicit diffusion
D_x	$N_x \times (N_x - 1)$	Space first-difference	Interior faces only
$L_x = D_x D_x^\top$	$N_x \times N_x$	Neumann Laplacian	$L_x \approx -\partial_{xx}$
A_ρ	$(N_t - 1) \times N_t$	Time avg: Grid 1→Grid 2	(5)
A_m	$(N_x - 1) \times N_x$	Space avg: Grid 1→Grid 2	(6)

For $N_t = 4$ the time operators are (after scaling by Δt):

$$D_t = \frac{1}{\Delta t}\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}, \qquad A_t = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

9.3 Right-Hand Side Vector d

The boundary conditions $\rho(0, \cdot) = \rho_0$, $\rho(1, \cdot) = \rho_1$ are absorbed into the right-hand side:

$$d = \frac{1}{\Delta t}\begin{pmatrix} \rho_0 \\ 0 \\ \vdots \\ 0 \\ -\rho_1 \end{pmatrix}, \tag{30}$$

with $d \in \mathbb{R}^{N_t \times N_x}$ (time-slow ordering). The factor $1/\Delta t$ comes from the first-difference operator D_t acting on a constant vector.

10 Algorithm Summary

Algorithm 1 Two-grid Linearized ADMM for the Schrödinger Bridge

Require: ρ_0, ρ_1: boundary densities; N_t, N_x, ε, γ, τ, α

Build operators D_t, A_t, D_x, L_x, A_ρ, A_m and RHS d
Initialize ρ by center-of-mass displacement interpolation of ρ_0
$m \leftarrow (\mu_1 - \mu_0)\,\rho$ (constant-velocity momentum)
$\bar{\rho} \leftarrow A_\rho\,\rho, \quad b \leftarrow A_m\,m, \quad \delta_{\bar{\rho}} \leftarrow 0, \quad \delta_b \leftarrow 0$
for $k = 1, 2, \ldots$ **do**
  Step 1: Proximal update
  $r_{\bar{\rho}} \leftarrow A_\rho \rho - \bar{\rho} + \delta_{\bar{\rho}}/\gamma, \quad r_b \leftarrow A_m m - b + \delta_b/\gamma$
  $u \leftarrow \rho - (\gamma/\tau)\,A_\rho^\top r_{\bar{\rho}}, \quad v \leftarrow m - (\gamma/\tau)\,A_m^\top r_b$
  $(\rho, m) \leftarrow \mathrm{prox}_{J_h/\tau}(u, v)$ (pointwise cubic solve, §6)
  Step 2: Projection update (with over-relaxation)
  $p_{\bar{\rho}} \leftarrow \alpha A_\rho \rho + (1-\alpha)\bar{\rho} + \delta_{\bar{\rho}}/\gamma$
  $p_b \leftarrow \alpha A_m m + (1-\alpha) b + \delta_b/\gamma$
  $(\bar{\rho}, b) \leftarrow \mathrm{proj}_C(p_{\bar{\rho}}, p_b)$ (DCT + tridiagonal solve, §8)
  Step 3: Dual update
  $\delta_{\bar{\rho}} \leftarrow \delta_{\bar{\rho}} + \gamma(p_{\bar{\rho}} - \bar{\rho}), \quad \delta_b \leftarrow \delta_b + \gamma(p_b - b)$
  Check primal/dual residuals; break if $<$ `tol`
end for

11 Convergence Diagnostics

At each iteration we track:

1. **Primal residual** (normalized): $r_p = \|A(\rho^k, m^k) - (\bar{\rho}^k, b^k)\| / \sqrt{N_{\text{primal}}}$
2. **Dual residual** (normalized): $r_d = \gamma\,\|A^\top((\bar{\rho}^k, b^k) - (\bar{\rho}^{k-1}, b^{k-1}))\| / \sqrt{N_{\text{dual}}}$
3. **Objective value**: $J_h(\rho^k, m^k) = \Delta t\,\Delta x \sum_{ij} m_{ij}^2/(2\rho_{ij})$
4. **Fokker–Planck violation**: $\|(I_x \otimes D_t - \varepsilon L_x \otimes A_t)\bar{\rho}^k + (D_x \otimes I_t) b^k - d\|$

Convergence is declared when $r_p <$ `tol` (default 10^{-4}). The FP violation is an independent check that the projection is working correctly; it should be near machine precision at every iteration.

12 Connection to the Literature

This formulation follows the line of work by Benamou and Brenier (2000) who introduced the dynamic OT problem (1)–(2), and the ADMM discretization approach of Papadakis,

Peyré, and Oudet (2014) who developed the proximal splitting on two grids for the OT case ($\varepsilon = 0$).

The extension to $\varepsilon > 0$ (Schrödinger Bridge) adds the diffusion term $\varepsilon L_x \otimes A_t$ to the Fokker–Planck constraint and modifies the M_{SB} operator accordingly. The semi-implicit treatment of this term via A_t (averaging in time) is the key stability contribution: it is the discrete analogue of the Crank–Nicolson scheme for the heat equation but embedded inside the ADMM projection step rather than an explicit time-stepper.

For Gaussian boundary conditions, the exact SB path has an analytic formula: $\rho(t, \cdot) = \mathcal{N}(\mu(t), \sigma^2(t))$ where $\mu(t)$ is linear in t and $\sigma^2(t)$ follows from the cross-covariance $c = -\varepsilon T + \sqrt{(\varepsilon T)^2 + \sigma_0^2 \sigma_1^2}$. This provides a ground-truth comparison for the numerical solver.

13 Parameters and Test Cases

13.1 Recommended Parameters

Table 2: Recommended parameter settings.

Parameter	Default	Notes
N_t	64	Time cells
N_x	128	Space cells
ε	0–0.05	0 = pure OT
γ	1.0	ADMM penalty
τ	5.0	Proximal parameter; need $\tau > \gamma \|A\|^2 \approx \gamma$
α	1.5	Over-relaxation (Boyd et al. 2011)
Tolerance	10^{-4}	On normalized primal residual

13.2 Test Cases

Test 1 — Gaussian → Gaussian, $\varepsilon = 0$ (OT): $\rho_0 = \mathcal{N}(0.25, 0.05^2)$, $\rho_1 = \mathcal{N}(0.75, 0.05^2)$. Expected objective: $J^* = \frac{1}{2}(0.75 - 0.25)^2 = 0.125$. The density evolves as a linear displacement interpolation.

Test 2 — Same marginals, $\varepsilon = 0.05$ (SB): Expect smoother, wider paths; $J_{\mathrm{SB}} > J_{\mathrm{OT}}$ due to the entropic cost. Comparison against the analytic Gaussian formula validates the solver.

Test 3 — Two humps → one hump: $\rho_0 = \frac{1}{2}\mathcal{N}(0.25, 0.05^2) + \frac{1}{2}\mathcal{N}(0.75, 0.05^2)$, $\rho_1 = \mathcal{N}(0.5, 0.1^2)$. Tests behavior with bimodal initial data.

A Why Not Discretize the HJB Equation?

An alternative approach is to derive the optimality conditions of the continuous problem first. The Pontryagin maximum principle yields a coupled system of a forward Fokker–

Planck equation for ρ and a backward Hamilton–Jacobi equation for the value function ϕ, linked through the optimal velocity $v^* = \partial_x \phi$. This *Schrödinger system* can be solved iteratively (e.g., Sinkhorn-like fixed-point iterations in the static setting).

The DtO approach chosen here has several practical advantages over this route:

1. **No adjoint PDE.** The ADMM objective encodes the constraints implicitly; there is no backward PDE to discretize consistently.

2. **Arbitrary boundary conditions.** Periodic, Neumann, and Dirichlet boundary conditions are handled identically by replacing the operators D_t, D_x, L_x without changing the algorithm.

3. **Direct access to (ρ, m).** The primal variables are the physically meaningful density and momentum, not potentials. Positivity of ρ is enforced at every iteration via the proximal operator.

4. **No CFL constraint.** As explained in Section 7, the implicit diffusion in the projection step sidesteps the parabolic stability restriction entirely.

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